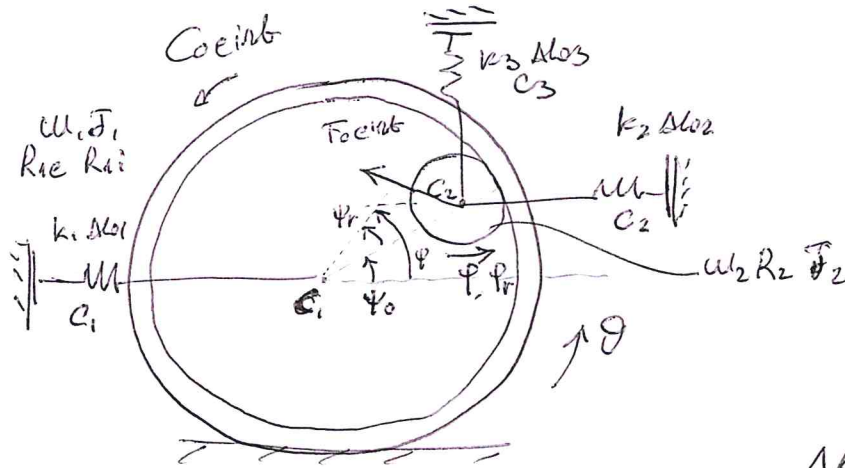


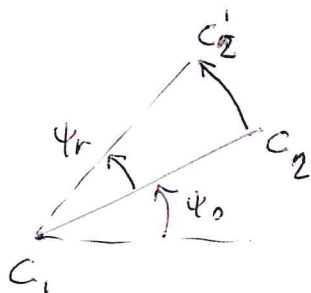


$$\Delta R = R_{1i} - R_2$$

- φ : absolute rotation of disk 2
 φ_r : relative " " " " with respect to ϑ : $\varphi_r = \varphi - \vartheta$
 ψ : absolute angle, measured with respect to the horizontal direction, of the line $\bar{C}_1\bar{C}_2$

$$1^{\circ} \text{ solution : } \underline{x} = \begin{Bmatrix} \vartheta \\ \varphi_r \end{Bmatrix}$$

φ_r : relative ~~rot~~ angle of the line $\bar{C}_1\bar{C}_2$ measured with respect to the direction of $\bar{C}_1\bar{C}_2$ in the static configuration



$$\rightarrow \varphi_r \Delta R = -\varphi_r R_2 \rightarrow \varphi_r = -\varphi_r \frac{R_2}{\Delta R}$$

$$\psi = \psi_0 + \varphi_r + \vartheta = \psi_0 + \vartheta - \varphi_r \frac{R_2}{\Delta R}$$

$$x_{C1} = -R_1 \vartheta$$

$$x_{C2} = -R_1 \vartheta + \Delta R \cos \psi = -R_1 \vartheta + \cos \left(\psi_0 + \vartheta - \varphi_r \frac{R_2}{\Delta R} \right)$$

$$y_{C2} = \Delta R \sin \psi = \Delta R \sin \left(\psi_0 + \vartheta - \varphi_r \frac{R_2}{\Delta R} \right)$$

$$\Delta x_1 = x_{C1} - x_{C1p} = -R_1 \vartheta$$

$$\Delta x_2 = -x_{C2} + x_{C2p}$$

$$\Delta x_3 = -y_{C2} + y_{C2p}$$

$$y_{C2} = y_{C2}$$

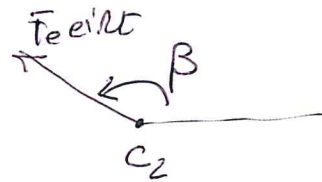
$$\varphi = \vartheta + \varphi_r$$

$$T = \frac{1}{2} m_1 \dot{x}_{c1}^2 + \frac{1}{2} J_1 \dot{\theta}^2 + \frac{1}{2} m_2 \dot{x}_{c2}^2 + \frac{1}{2} m_2 \dot{y}_{c2}^2 + \frac{1}{2} J_2 \dot{\phi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^3 c_i \Delta l_i^2, \quad V_{\text{el}} = \frac{1}{2} \sum_{k=1}^3 k_i \Delta l_i^2, \quad V_g = m_2 g h_{c2}$$

$$SW = C e^{i\alpha t} \delta\theta + \bar{F}_0 \cos\beta e^{i\alpha t} \delta x_{c2} + \bar{F}_0 \sin\beta e^{i\alpha t} \delta y_{c2}$$

first derivatives $\rightarrow$



$$\underline{A}_m = \begin{bmatrix} -R_1 e & \phi \\ 1 & \phi \\ -R_1 e - \Delta R \sin \phi_0 & R_2 \sin \phi_0 \\ \Delta R \cos \phi_0 & -R_2 \cos \phi_0 \\ 1 & 1 \end{bmatrix}$$

$$\underline{A}_{\Delta R} = \begin{bmatrix} -R_1 e & \phi \\ R_1 e + \Delta R \sin \phi_0 & -R_2 \sin \phi_0 \\ -\Delta R \cos \phi_0 & +R_2 \cos \phi_0 \end{bmatrix}$$

$\underline{A}_c = \underline{A}_{\Delta R}$

$$(\underline{A}_g) = \begin{bmatrix} 1 & 0 \\ -R_1 e - \Delta R \sin \phi_0 & R_2 \sin \phi_0 \\ \Delta R \cos \phi_0 & -R_2 \cos \phi_0 \end{bmatrix}$$

$$\underline{F} = \begin{bmatrix} C e^{i\alpha t} \\ \bar{F}_0 \cos\beta e^{i\alpha t} \\ \bar{F}_0 \sin\beta e^{i\alpha t} \end{bmatrix}$$

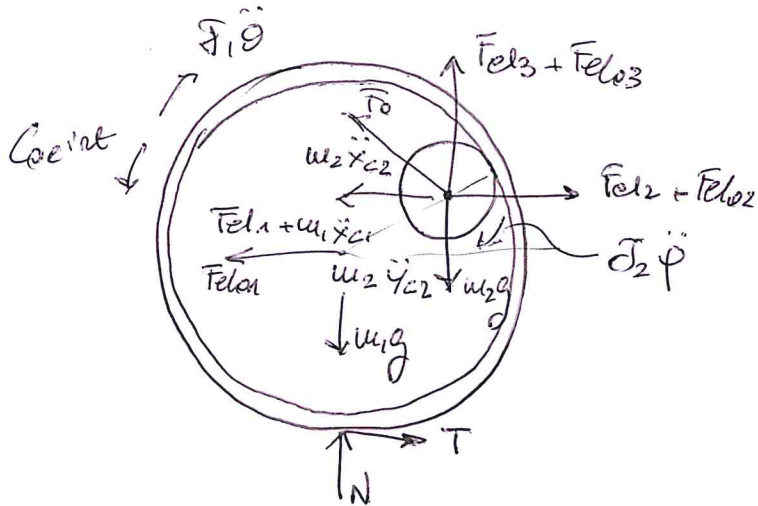
second derivatives $\rightarrow$

$$[k_{\Delta R} e_2] = \begin{bmatrix} \Delta R \cos \phi_0 & -R_2 \cos \phi_0 \\ -R_2 \cos \phi_0 & \frac{R_2^2}{\Delta R} \cos \phi_0 \end{bmatrix} k_2 \Delta l_02$$

$$[k_{\Delta R} e_3] = \begin{bmatrix} \Delta R \sin \phi_0 & -R_2 \sin \phi_0 \\ -R_2 \sin \phi_0 & \frac{R_2^2}{\Delta R} \sin \phi_0 \end{bmatrix} k_3 \Delta l_03$$

$$[k_{g2}] = \begin{bmatrix} -\Delta R \sin \phi_0 & +R_2 \sin \phi_0 \\ R_2 \sin \phi_0 & -\frac{R_2^2}{\Delta R} \sin \phi_0 \end{bmatrix} m_2 g$$

question nr. 5.1



$$\sum \bar{F}_x = 0 \rightarrow T - \bar{F}_{el1} - m_1 \ddot{x}_{c1} + \bar{F}_{el2} - m_2 \ddot{x}_{c2} + \bar{F}_0 \cos \beta = 0 \quad \bar{F}_0 = 0$$

$$T_{ST} = \bar{F}_{el1} - \bar{F}_{el2}$$

$$\boxed{\bar{F}_{dyn} = \bar{F}_{el1} + m_1 \ddot{x}_{c1} - \bar{F}_{el2} + m_2 \ddot{x}_{c2}}$$

$$\bar{F}_y = 0 \rightarrow N - m_1 g - m_2 g + \bar{F}_{el3} + \bar{F}_{el03} - m_2 \ddot{y}_{c2} + \bar{F}_0 \sin \beta = 0$$

$$N_{ST} = m_1 g + m_2 g - \bar{F}_{el03}$$

$$\boxed{N_{dyn} = m_2 \ddot{y}_{c2} - \bar{F}_{el3}}$$

loss of contact when $N = N_{dyn} + N_{ST} = 0 \rightarrow |N_{dyn}| = N_{ST}$

$|N_{dyn}|$: amplitude computed with Coil at $f = 2 \text{ Hz}$

$$\bar{C}_0 = C_0 \frac{|N_{dyn}|}{N_{ST}}$$

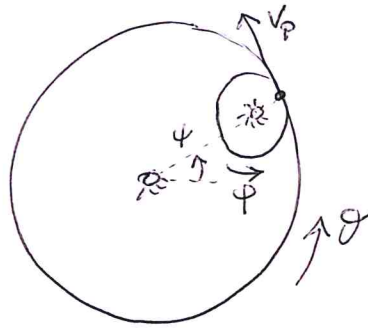
it's a linear problem $\rightarrow$ just a proportion between two cases.

2nd solution

$$\underline{x} = \begin{Bmatrix} \theta \\ \psi \end{Bmatrix}$$

computation of the rotation ψ

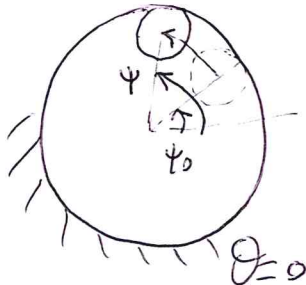
- 1) $\theta \neq 0$ $\psi = \psi_0 = \text{const}$ the direction $\vec{C_1 C_2}$ is fixed because $\psi = \psi_0$. Not considering the translation - Ried of point C_1 :



v_p is the same because of the rolling without sliding condition $\rightarrow$

$$\begin{aligned} v_p &= \dot{\theta} R_{1i} \\ v_p &= \dot{\psi} R_2 \rightarrow \dot{\psi} = \dot{\theta} \frac{R_{1i}}{R_2} \end{aligned}$$

- 2) $\theta \neq 0$ $\psi \neq \psi_0$



$$(\psi - \psi_0) \Delta R = -\dot{\psi}_r R_2$$

$$\downarrow$$

$$\dot{\psi} = \dot{\theta} \frac{R_{1i}}{R_2} - (\psi - \psi_0) \frac{\Delta R}{R_2}$$

$$\dot{\psi}_r = \dot{\psi} - \dot{\theta} \quad (\text{question 4})$$

$$x_{c1} = -R_1 \theta$$

$$x_{c2} = -R_1 \theta + \Delta R \cos \psi$$

$$y_{c2} = \Delta R \sin \psi$$

$$\begin{aligned} \Delta L_1 \\ \Delta L_2 \\ \Delta L_3 \end{aligned} \left. \vphantom{\begin{aligned} \Delta L_1 \\ \Delta L_2 \\ \Delta L_3 \end{aligned}} \right\} \text{ same as previous solution}$$

first and second derivatives $\rightarrow$

$$\Delta u = \begin{bmatrix} -R\dot{e} & \phi \\ 1 & \phi \\ -R\dot{e} & -\Delta R \sin \phi_0 \\ 0 & \Delta R \cos \phi_0 \\ \frac{R\dot{e}}{R_2} & -\frac{\Delta R}{R_2} \end{bmatrix}$$

$$\Delta h = \Delta c = \begin{bmatrix} -R\dot{e} & \phi \\ R\dot{e} & \Delta R \sin \phi_0 \\ 0 & -\Delta R \cos \phi_0 \end{bmatrix}$$

$$(\Delta e) = \begin{bmatrix} 1 & 0 \\ -R\dot{e} & -\Delta R \sin \phi_0 \\ 0 & \Delta R \cos \phi_0 \end{bmatrix}$$

$$[k_{\Delta e 2}] = k_2 \Delta h_2 \begin{bmatrix} 0 & 0 \\ 0 & \Delta R \cos \phi_0 \end{bmatrix}$$

$$[k_{\Delta e 3}] = k_3 \Delta h_3 \begin{bmatrix} 0 & 0 \\ 0 & \Delta R \sin \phi_0 \end{bmatrix}$$

$$[k_{s2}] = u_2 g \begin{bmatrix} 0 & 0 \\ 0 & -\sin \phi_0 = \Delta R \end{bmatrix}$$